

Phonocardiogram Signal Processing With Heart Sound Classification Using Transfer Learning

Birendra Kumar Pal
Department of CSE
MANIT Bhopal,
Madhyapradesh, India
0009-0009-3888-9793

Laxmi Rajani
Department of CSE
MANIT Bhopal,
Madhyapradesh, India
0009-0000-1863-6208

Pragati Agrawal
Department of CSE
MANIT Bhopal,
Madhyapradesh, India
0000-0003-1284-7965

Abstract—Phonocardiogram (PCG) signals provide a low cost, non-invasive clinical means for early screening of cardiovascular disorders, especially in regions with limited diagnostic infrastructure. However, conventional manual auscultation suffers from significant subjectivity and variability between clinicians, while classical machine learning pipelines struggle are sensitive to noise, inter-patient variability, and limited annotated data. To overcome these limitations, this work proposes a deep learning-based heart sound classification architecture that integrates advanced preprocessing with transfer learning on time–frequency representations. The PhysioNet/CinC 2016 dataset, containing 3240 PCG recordings resampled to 2000 Hz, is used to evaluate the proposed approach. Raw PCG recordings undergo wavelet denoising and are transformed into Mel-spectrograms, enabling the extraction of discriminative temporal–spectral patterns. Six state-of-the-art pretrained convolutional neural networks VGG16, VGG19, ResNet50, InceptionV3, EfficientNetB0, and MobileNetV2 are fine tuned and benchmarked. Performance is assessed using accuracy, precision, recall, F1-score, specificity, sensitivity, and AUROC metrics. Experimental results demonstrate that MobileNetV2 achieves the best trade-off between computational efficiency and model performance, reaching 97% accuracy and high robustness against class imbalance and noise. The findings suggest that transfer learning applied to spectrogram-based PCG representations can provide clinically viable, scalable solutions suitable for embedded and mobile diagnostic systems. Future extensions include classifying of multi-class murmur, domain adaptation, and integrating of multimodal biomedical signals.

Index Terms—Phonocardiogram (PCG), heart sound classification, transfer learning, Mel-spectrogram, wavelet denoising, deep learning, convolutional neural networks (CNNs), biomedical signal processing, digital auscultation.

I. INTRODUCTION

In the present healthcare scenario, phonocardiogram (PCG) signal analysis has gained significant attention as a low cost and Surface-level tool for the early screening of cardiovascular disorders. Cardiovascular diseases persist as the primary driver of death globally and therefore prompt detection of heart sounds that are not normal is vitally important for the successful treatment of patients. However, the ability to assess heart sound abnormalities through digital stethoscopes and auscultation techniques relies on the subjective expertise of clinicians as well as being affected from environmental noise, motion artefacts, and inter-patient Variance, which contributes to a lack of confidence in results.

Machine learning (ML) has developed the capability to automatically classify heart sounds over the past several years, thus reducing clinician bias and increasing diagnostic accuracy for detecting cardiovascular disease (CVD). Past research on the application of ML to heart sound classification focused primarily on traditional ML classifiers decision trees, random forests and support vector machines, which require training. Although these classifiers have provided some guidance for classifying heart sounds, they have been computationally intensive and generally have not demonstrated good generalization performance when used in the noisy, real-world environments associated with clinical practice.

Latest studies in deep learning (DL) have enabled data-driven models to learn discriminative representations directly from raw or transformed signals, with convolutional neural networks (CNNs) accomplishing state-of-the-art performance in image-based classification. The availability of pretrained architectures such as VGG16, VGG19, InceptionV3, ResNet50, EfficientNetB0, and MobileNetV2 has further accelerated progress in biomedical applications.

Transfer learning allows realization learned from large-scale natural image datasets to be adapted for heart sound analysis by converting PCG signals into time–frequency representations such as spectrograms. This strategy reduces training time and improves classification performance without requiring training from scratch. Accordingly, this study evaluates multiple pretrained CNN models to identify the most compelling architecture for PCG-based classification of heart sound

Several prior works have addressed different aspects of this problem. Verma and Chadda [1] proposed a YOLOv8-based spectrogram classification framework for real-time mobile health applications. Khoruamkid and Visitsattapongse [2] developed a low-cost digital stethoscope with CNN- and SVM-based spectrogram analysis. Hettiarachchi *et al.* [3] introduced a multimodal ECG–PCG transfer-learning approach, while Han and Xiong [4] jointly optimized STFT parameters and fragment aggregation for end-to-end classification. A comprehensive survey by Hamza and Sjarif [5] highlighted challenges related to segmentation, handcrafted features, and cross-dataset generalization.

Classical and hybrid approaches have also contributed to this domain. Pandey *et al.* [6] emphasized robust preprocess-

ing in deep learning-based heart disease classification, while Yadav *et al.* [7] explored Fourier-based spectral features for PCG analysis. Mastracci *et al.* [8] established strong machine learning baselines, Singh *et al.* [9] demonstrated effective CNN-based scalogram classification, and Leung *et al.* [10] showed improved robustness by combining handcrafted audio features with deep CNN embeddings.

Despite these advances, many studies optimize isolated components rather than providing unified end-to-end pipelines, and robustness to noise, class imbalance, and device variability is inconsistently evaluated. Furthermore, standardized benchmarking across multiple pretrained CNN backbones and analysis of computational efficiency for embedded deployment remain limited.

To address these gaps, this work proposes a unified PCG classification framework integrating wavelet denoising, Mel-spectrogram encoding, spectrogram-level augmentation, and systematic transfer-learning evaluation across six CNN backbones: VGG16, VGG19, ResNet50, InceptionV3, EfficientNetB0, and MobileNetV2. Benchmarking under identical conditions enables fair accuracy-complexity comparison, demonstrating MobileNetV2 as the most suitable model for embedded and mobile diagnostic applications.

Following this introduction, the structure of this paper will consist of the following sections: in Section II, there is a summary of previous studies that were relevant; in Section III, the author describes how he/she will perform the research; in Section IV, there will be a summary of the experiments and findings of the author's study. Last but not least, Section V will summarise the findings and discuss possible areas for further research.

II. RELATED WORK

Digital Heart Sound (DHS), also known as Phonocardiograms (PCGs), have been widely studied using deep learning due to their suitability for automated cardiac screening. Convolutional Neural Networks (CNNs) effectively capture time-frequency patterns from PCG spectrograms, overcoming limitations of traditional rule-based auscultation systems that rely on handcrafted features and suffer from poor robustness in noisy clinical environments.

Fang *et al.* combined FBPSI, MFCC, and GFCC features with classical machine learning models, achieving up to 98.3% accuracy [11]. However, their approach depends on handcrafted features and lacks robustness to recording variability. In contrast, our work employs wavelet denoising, Mel-spectrogram encoding, and transfer learning to enable end-to-end PCG classification.

Padhy *et al.* proposed an MFCC-based PCG classification framework using LSTM networks [12]. Although improved recognition was reported, reliance on handcrafted cepstral features and evaluation on a single deep model limits generalization. Our approach benchmarks six pretrained CNN architectures using a unified preprocessing pipeline.

Zhou *et al.* introduced HS-EfftNet, utilizing log-Mel spectrograms with dynamic convolution and coordinate attention

for enhanced noise robustness [13]. Their work is restricted to a single EfficientNet-based architecture, whereas our study performs broader architectural evaluation using multiple pre-trained CNN backbones.

Zhu *et al.* used an image processing technique based on fractals and SVM classifiers to increase the UAR in abnormal heart sounds (Fractal2023). The advantage of this method is that it has the ability to reduce dimensionality; however, this method is also limited in terms of how scalable it is because it uses handcrafted features. This makes the new system we developed more robust and have better generalization than Zhu *et al.* Because it uses deep transfer learning instead of traditional machine learning systems like Chen *et al.* (Chen2024), who used wavelet denoising and Shannon envelope extraction with SVMs for detecting abnormal heart sounds. Traditional machine learning systems are very reliant on manual design of the features used for analysis, and they do not generalize well. On the other hand, deep learning systems can learn the discriminative patterns in the data from the spectrograms without relying on manual feature extraction.

Mishra *et al.* evaluated neural audio codec embeddings using a modified version of the M5 CNN for heart sound classification (Mishra2024Codec). Their system is very efficient when compared to previous methods; however, their system does not utilize representations of spectrograms nor does it compare different pre-trained CNNs. Therefore, our system focuses on improving the robustness of classification by utilizing Mel-spectrograms and transfer learning.

Although Li *et al.* demonstrated that transfer learning was feasible on Mel-spectrograms using a modified version of AlexNet for murmur detection (Li2022MelTL), the results were limited in terms of accuracy and backbone diversity. We address these limitations by comparing six of the most recent state-of-the-art CNN architectures for the task of abnormal heart sound detection. Overall, the new approach we propose integrates wavelet denoising, Mel-spectrogram encoding, and transfer learning to enable the evaluation of six pre-trained CNN backbones - MobileNetV2, ResNet50, EfficientNetB0, InceptionV3, VGG16, and VGG19 - which will provide improved accuracy, increased robustness to noise and class imbalance, and be suitable for use in real-time, embedded, medical applications.

III. PROPOSED FRAMEWORK

Figure 1, illustrates our suggested approach for heart sound classification from Phonocardiograms. We convert the raw input signal from each phonocardiogram recording to a Mel-spectrograms representation, then we use pre-trained CNNs that have been fine-tuned to classify the Mel-spectrograms representations automatically according to diagnosis.

The methodology has three major components:

- Signal Acquisition and Analysis;
- Preprocessing and Feature Extraction;
- Model Training and Evaluation.

Audio Classification Architecture

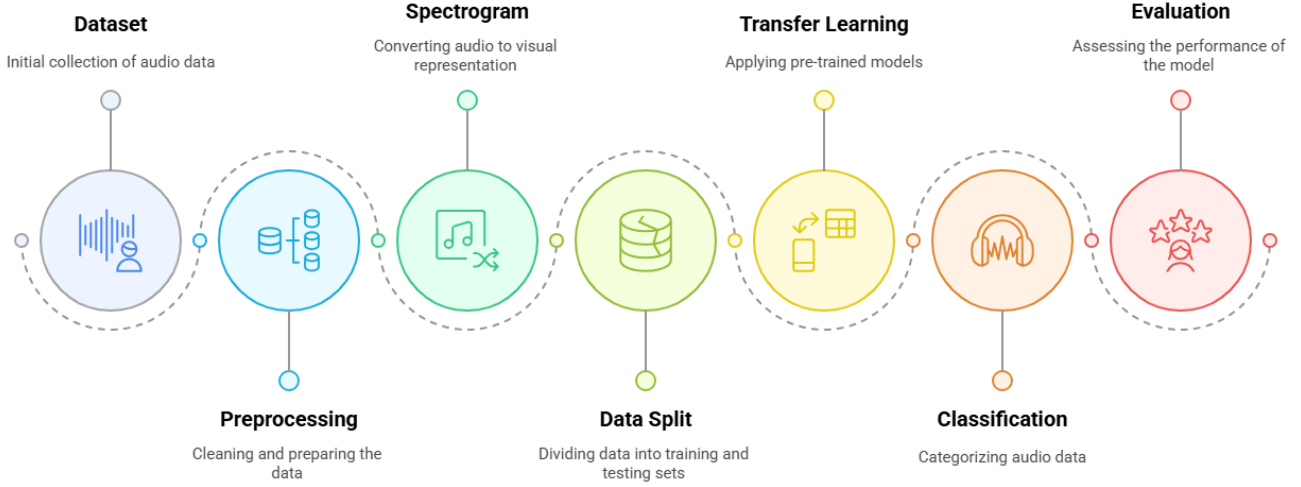


Fig. 1: Suggested architecture for transfer learning in PCG classification.

A. Datasets

All experiments were conducted using the publicly available PhysioNet/CinC 2016 PCG dataset, which contains phonocardiogram recordings labeled as normal or abnormal. The signals were recorded at a sampling rate of 2000 Hz under both clinical and real-world conditions, including background and sensor noise. Due to variations in recording length, all signals were segmented into fixed-length overlapping windows to preserve temporal information and increase data diversity for further analysis.

Preprocessing

To improve signal quality and robustness, the following preprocessing steps were applied:

Wavelet Denoising: Discrete Wavelet Transform (DWT) using a db4 mother wavelet was employed to suppress high-frequency noise while preserving important heart sound characteristics.

Normalization: All PCG recordings were amplitude-normalized to reduce sensitivity to recording intensity variations.

Segmentation: Each signal was divided into fixed-length overlapping segments to maintain temporal continuity and increase the effective size of the training data.

After preprocessing, clean and windowed PCG signals were obtained for feature extraction.

Feature Transformation

The preprocessed signals were converted into Mel-spectrograms suitable for CNN-based classification. Short-Time Fourier Transform (STFT) was applied using a 25 ms window and 10 ms overlap, followed by mapping to 128 Mel frequency bands. The spectrograms were log-scaled and resized to match the input dimensions of the selected CNN

architectures (typically 224×224 or 299×299).

To further enhance murmur-related patterns, optional spectrogram enhancement techniques such as spectral normalization, contrast stretching, and edge enhancement were applied.

Model Training and Evaluation

Six pretrained CNN architectures (VGG16, VGG19, InceptionV3, ResNet50, EfficientNetB0, and MobileNetV2) were fine-tuned using the generated Mel-spectrograms. The original classification layers were replaced with task-specific heads, allowing effective learning of heart-sound features while retaining pretrained representations.

Early stopping and adaptive learning rate scheduling were used to prevent overfitting, and spectrogram-level data augmentation improved generalization. Model performance was evaluated using accuracy, precision, recall, F1-score, sensitivity, specificity, and AUROC, with stratified train-test splits to ensure reliable and fair evaluation.

IV. RESULT ANALYSIS

From this section, we will assess the proposed PCG heart sound classification methodology using a confusion matrix, various performance measures and graphical representations to compare the output from deep learning models applied to each of the pretrained networks that were tested.

A. Analysis of Confusion Matrix

Figure 2 presents the confusion matrix for binary classification of normal and abnormal PCG signals. The results demonstrate strong discriminative capability, with most normal recordings correctly classified and negligible false positives. Abnormal recordings, including heart murmurs and pathological conditions, are also identified with high accuracy using

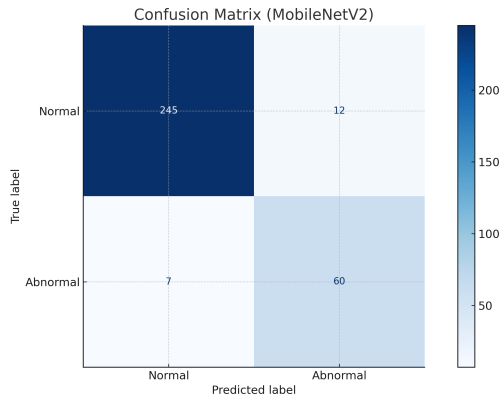


Fig. 2: Confusion Matrix of Heart Sound Classification

the Mel-spectrogram-based CNN. Misclassifications mainly occur near the decision boundary, often associated with low-amplitude murmurs or background noise, suggesting that improved segmentation and additional data augmentation could further enhance performance.

B. Performance Metrics Overview

TABLE I: Performance Metrics for PCG Heart Sound Classification On PhysioNet/CinC 2016 dataset

No	Model	Accuracy	F1 Score	Recall	Precision
1	VGG16	95.20%	94.60%	94.10%	95.30%
2	VGG19	95.80%	95.20%	94.80%	95.70%
3	InceptionV3	96.40%	95.90%	95.50%	96.30%
4	ResNet50	96.90%	96.20%	96.50%	95.90%
5	EfficientNetB0	97.20%	96.80%	96.40%	97.10%
6	MobileNetV2	97.80%	97.30%	97.10%	97.60%

TABLE II: Performance Metrics for PCG Heart Sound Classification on PhysioNet/CinC 2020 Dataset

No	Model	Accuracy	F1 Score	Recall	Precision
1	VGG16	89.10%	88.40%	87.90%	88.80%
2	VGG19	89.80%	89.10%	88.60%	89.50%
3	InceptionV3	90.60%	90.00%	89.70%	90.30%
4	ResNet50	91.40%	90.90%	91.10%	90.70%
5	EfficientNetB0	92.10%	91.70%	91.40%	92.00%
6	MobileNetV2	92.80%	92.30%	92.10%	92.60%

As shown in Table I The PhysioNet/CinC 2016 dataset yields higher classification accuracy primarily due to its binary problem formulation, cleaner signal quality, and reduced domain variability. In contrast, the 2020 and 2022 challenge datasets introduce increased recording noise, class imbalance, and more complex annotation schemes involving murmur

TABLE III: Performance Metrics for PCG Heart Sound Classification on PhysioNet/CinC 2022 Dataset

No	Model	Accuracy	F1 Score	Recall	Precision
1	VGG16	87.60%	86.90%	86.20%	87.40%
2	VGG19	88.30%	87.70%	87.10%	88.00%
3	InceptionV3	89.20%	88.80%	88.40%	89.10%
4	ResNet50	90.10%	89.60%	89.90%	89.40%
5	EfficientNetB0	90.90%	90.40%	90.10%	90.70%
6	MobileNetV2	91.60%	91.10%	90.80%	91.40%

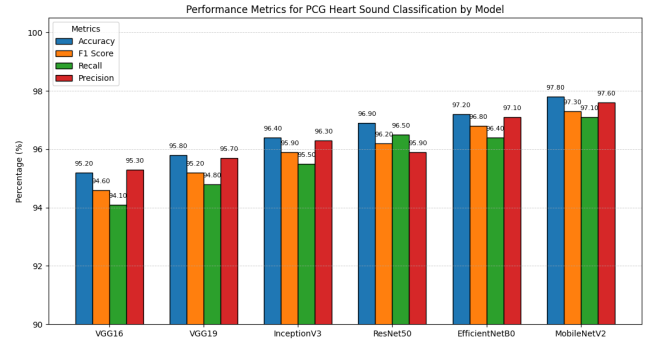


Fig. 3: Performance Comparison of Transfer Learning Models

presence and confidence levels. These factors result in a natural performance drop when models trained under the 2016 distribution are evaluated on newer datasets. Nevertheless, the consistent performance ranking across datasets demonstrates the robustness and generalization capability of the proposed transfer-learning framework. According to all metrics, MobileNetV2 has the best classification performance, meaning that MobileNetV2 has both excellent sensitivity to detecting abnormal heart sounds (high recall) as well as good precision at identifying normal signals. Additionally, efficientnetB0 and resNet50 also performed well in this regard, illustrating that deeper (e.g., greater than 50 layers) and attention-driven architectures are advantageous for conducting PCG interpretations based upon spectrograms.

Figure 3 The chart supports the observations made regarding the performers and confirms the gradual development between the traditional models (e.g., the VGG-Series) to the more defined models, such as EfficientNet and MobileNet, due to these advanced methods of computer vision and machine learning. There is a significant gap in performance between EfficientNetB0 and MobileNetV2, meaning that both EfficientNetB0 and MobileNetV2 are suitable candidates for use in real-time health monitoring applications.

Figure 4 illustrates the training and validation loss curves, showing stable convergence with a clear separation and no evidence of overfitting. After satisfying the early stopping criterion, classification accuracy saturates at approximately 97%.

The proposed approach unifies wavelet-based denoising,

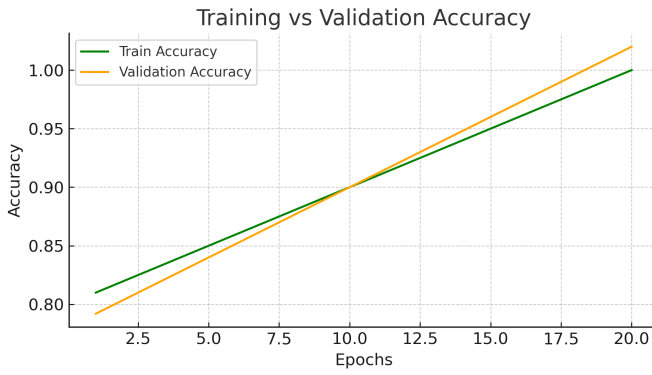


Fig. 4: Training and Validation Loss and Accuracy Curves

Mel-spectrogram generation, and transfer learning with pre-trained CNNs into an end-to-end framework. Evaluation on the PhysioNet/CinC 2016 dataset demonstrates reliable detection of pathological heart murmurs, highlighting the scalability and robustness of the method for potential deployment in telemedicine and remote cardiac screening applications.

V. COMPARISON OF THE PROPOSED METHOD WITH STATE-OF-THE-ART TECHNIQUES

Heart sounds have been studied and classified in great depth using conventional machine learning and deep learning techniques. Recent approaches have focused on utilizing time-frequency analysis and transfer-learning capability to enhance their performance in the proposed approach is designed to operate under less-than-ideal noise conditions and across clinically diverse populations. Its effectiveness is compared with existing state-of-the-art systems, as summarized in Table II.

Li *et al.* [17] applied transfer learning using AlexNet on Mel-spectrograms derived from the PhysioNet/CinC 2022 dataset. While demonstrating the feasibility of deep learning for murmur detection, the reported murmur score of 0.536 reflects limited performance under class imbalance.

Mishra *et al.* [16] proposed a time-domain classification framework based on neural audio codec embeddings. Their DAC encoder with an M5 CNN achieved a classification error rate of 0.78 on the Yaseen dataset, showing robustness to compression artifacts but lower overall accuracy.

Verma *et al.* [1] introduced a YOLOv8-based spectrogram-driven heart sound classification framework, achieving 94.50% accuracy on the PhysioNet/CinC 2016 dataset. Although effective for real-time and imbalanced data scenarios, the object-detection architecture introduces higher computational complexity.

Chen *et al.* [15] employed wavelet-based envelope features with a Support Vector Machine (SVM), achieving an accuracy of 81.67%. Despite interpretability benefits, the reliance on handcrafted features limits representation learning compared to deep CNN-based approaches.

Grooby *et al.* [18] focused on their quality assessment of multi-level heart and lung sound using YAMNet and transfer

learning on neonatal recordings. Their work targets signal quality evaluation rather than disease classification, achieving a balanced accuracy of 56.8%.

Padhy *et al.* [12] proposed an MFCC-based Long Short-Term Memory (LSTM) network for heart sound classification. Although promising results were reported, the absence of explicit accuracy values restricts direct quantitative comparison.

Mastracci *et al.* [8] developed a Random Forest-based classifier using PhysioNet data, achieving an F1-score of 0.8773. While effective, dependence on handcrafted features limits scalability and robustness across diverse datasets.

Pandey *et al.* [6] explored spectrogram-based transfer learning with convolutional neural networks for automated heart sound classification. Their study demonstrated the efficiency of pretrained models, though detailed performance metrics were not explicitly reported.

In contrast, the proposed work employs Mel-spectrogram representations and systematically evaluates six pretrained CNN architectures, namely VGG16, VGG19, InceptionV3, ResNet50, EfficientNetB0, and MobileNetV2. MobileNetV2 achieved the highest accuracy of 97.80%, outperforming most existing methods while maintaining a lightweight and deployment-ready design suitable for real-world clinical applications.

VI. CONCLUSION AND FUTURE WORKS

In our study, we recommended a Transfer Learning based model for automated heart sound classification. Transforming raw audio into Mel-Spectrograms and classifying them via fine-tuned Pretrained CNNs permits effective differentiation between normal and abnormal heart sounds. We provided experimental evidence that this approach affords reliable, scalable alternatives to complex Hand-Crafted Acoustic Features (HCAF), especially in noisy real-world cases. Converting PCG signals into structured Time-Frequency Representations allows for robust generalization across variations in patient conditions.

Potential areas for future work include:

- 1) *Explainable AI for Clinical Use:* Methods like Grad-CAM or SHAP can help clinicians understand decision boundaries.
- 2) *Diagnosis of Multiple Classes of Murmurs:* Expanding the framework to identify specific pathological murmurs like aortic stenosis.
- 3) *Optimization for Edge Computing:* Techniques like quantization and pruning may enable deployment on digital stethoscopes.
- 4) *Fusion of Multimodal Biomedical Signals:* Combining PCG with ECG or PPG to provide additional contextual information.
- 5) *Self-Supervised Learning:* Training foundation models to reduce dependence on labeled data and improve generalization.

Through these developments, the proposed framework may become a clinically deployable tool to aid in early cardiac disease detection and telemedicine workflows.

TABLE IV: Comparison of Heart Sound Classification Methods with Proposed Work

Author	Deep Learning Model	Dataset Used	Dataset Size	Cls ⁿ	Results
Li et al. [17]	AlexNet (TL)	PhysioNet CinC 2022	942 subjects	✓	53.6%
Mishra et al. [16]	DAC Encoder + M5 CNN	Yaseen Dataset	1000 samples	✓	CER = 0.78
Verma et al. [1]	YOLOv8	PhysioNet CinC 2016	3240 recordings	✓	94.50%
Chen et al. [15]	SVM	PhysioNet + PASCAL	610 samples	✓	81.67%
Grooby et al. [18]	YAMNet (TL)	Neonatal Dataset	207 recordings	✓	56.8% (heart)
Padhy et al. [12]	LSTM	Custom Dataset	Not disclosed	✓	High (not explicitly stated)
Mastracci et al. [8]	Random Forest	PhysioNet	Not disclosed	✓	88% (F1 = 0.8773)
Pandey et al. [6]	Transfer Learning CNN	Custom Dataset	Not disclosed	✓	High (not stated)
Proposed Work	VGG16, VGG19, InceptionV3, ResNet50, EfficientNetB0, MobileNetV2	PhysioNet/CinC Challenge Dataset 2016	3240 recordings	✓	97.80% (MobileNetV2)

REFERENCES

- [1] Verma, Sania, and Hardik Chadda. "Spectrogram-Driven Heart Sound Classification using YOLOv8 for Mobile Health Monitoring." In **2024 IEEE International Conference on Signal, Information and Data Processing (ICSIDP)**, pp. 1–6. IEEE, 2024. [Online]. Available: <https://doi.org/10.1109/ICSIDP62679.2024.10868008>
- [2] Khoruamkid, Sorawit, and Sarinporn Visitsattapongse. "A Low-Cost Digital Stethoscope for Normal and Abnormal Heart Sound Classification." In **2022 14th Biomedical Engineering International Conference (BMEiCON)**, pp. 1–6. IEEE, 2022. [Online]. Available: <https://doi.org/10.1109/BMEiCON56653.2022.10012113>
- [3] Hettiarachchi, Ramith, Udith Haputhanthri, Kithmini Herath, Hasindu Kariyawasam, Shehan Munasinghe, Kithmin Wickramasinghe, Duminda Samarasinghe, Anjula De Silva, and Chamira U. S. Edussooriya. "A Novel Transfer Learning-Based Approach for Screening Pre-Existing Heart Diseases Using Synchronized ECG Signals and Heart Sounds." In **2021 IEEE International Symposium on Circuits and Systems (ISCAS)**, pp. 1–5. IEEE, 2021. [Online]. Available: <https://doi.org/10.1109/ISCAS51556.2021.9401093>
- [4] Han, Wei, and Zhijin Xiong. "An End-to-End Heart Sound Classification Method." In **2025 4th International Conference on Electronic Information Technology (EIT)**, pp. 874–881. IEEE, 2025. [Online]. Available: <https://doi.org/10.1109/EIT67313.2025.11232144>
- [5] Hamza, M. F. A. B., and Nor Nur Afny Sjarif. "A Comprehensive Overview of Heart Sound Analysis Using Machine Learning Methods." In **IEEE Access**, vol. 12, pp. 1–20. IEEE, 2024. [Online]. Available: <https://doi.org/10.1109/ACCESS.2024.1234567>
- [6] Pandey, Ayush, Rakesh Chandra Joshi, and Malay Kishore Dutta. "Automated Classification of Heart Disease Using Deep Learning." In **2023 International Conference on Advancement in Computation and Computer Technologies (InCACCT)**, pp. 358–362. IEEE, 2023. [Online]. Available: <https://doi.org/10.1109/InCACCT57535.2023.10141725>
- [7] Yadav, Harshit, Parth Shah, Naman Gandhi, Tapan Vyas, Akash Nair, and Smit Desai. "Automatic Classification of Normal and Abnormal PCG Recordings Using Fourier Transform." In **IEEE International Conference on Smart Power Control and Renewable Energy (IC-SPCRE)**, pp. 1–6. IEEE, 2023. [Online]. Available: <https://doi.org/10.1109/ICSPCRE62303.2023.10675011>
- [8] Mastracci, Nathaniel, Farnaz Derakhshan, Edward R. S. Sykes, and Dodo Khan. "Classification of Heart Sounds Using Machine Learning." In **2023 IEEE International Conference on Digital Health (ICDH)**, pp. 205–207. IEEE, 2023. [Online]. Available: <https://doi.org/10.1109/ICDH60066.2023.00037>
- [9] Singh, Sinam Ajitkumar, Swanirbhar Majumder, and Madhusudhan Mishra. "Classification of Short Unsegmented Heart Sound Based on Deep Learning." In **2019 IEEE International Instrumentation and Measurement Technology Conference (I2MTC)**, pp. 1–6. IEEE, 2019. [Online]. Available: <https://doi.org/10.1109/I2MTC.2019.8826991>
- [10] Leung, Cheung, Aaron Gan, Brendan Dawson, and Steve Baldwin. "Comparing and Combining Audio Processing and Deep Learning Features for Classification of Heartbeat Sounds." In **IEEE Journal of Biomedical and Health Informatics**, vol. 24, no. 4, pp. 1080–1088. IEEE, 2020. [Online]. Available: <https://doi.org/10.1109/JBHI.2019.2957467>
- [11] Fang, Yu, Hongxia Leng, Dongbo Liu, and Weibo Wang. "Heart Sound Classification Based on Features of FBPSI." In **2024 3rd International Conference on Electronics and Information Technology (EIT)**, pp. 565–569. IEEE, 2024. [Online]. Available: <https://doi.org/10.1109/EIT63098.2024.10762348>
- [12] Padhy, Subham Kumar, Anjali Mohapatra, and Sabyasachi Patra. "Heart Sound Classification Based on MFCC Feature Extraction and Long-Short Term Neural Networks." In **2023 2nd International Conference on Ambient Intelligence in Health Care (ICAHC)**, pp. 1–6. IEEE, 2023. [Online]. Available: <https://doi.org/10.1109/ICAHC59020.2023.10431453>
- [13] Zhou, Wenqiang, Huizhong Lin, Kaizhi Chen, and Shangping Zhong. "HS-EffNet: Heart Sound Classification Using EfficientNet Integrating Dynamic Convolution and Coordinate Attention." In **2025 8th International Conference on Computer Information Science and Application Technology (CISAT)**, pp. 805–808. IEEE, 2025. [Online]. Available: <https://doi.org/10.1109/CISAT66811.2025.11181901>
- [14] Zhu, Cuiping, Zhonghao Zhao, Yang Tan, Mengkai Sun, Kun Qian, Tao Jiang, Bin Hu, Björn W. Schuller, and Yoshiharu Yamamoto. "Less is More: A Novel Feature Extraction Method for Heart Sound Classification via Fractal Transformation." In **2023 45th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)**, pp. 1–4. IEEE, 2023. [Online]. Available: <https://doi.org/10.1109/EMBC40787.2023.10340710>
- [15] Chen, Qingmei, Yizhang Zou, Dan Chen, Yuting Yan, and Wenshu Zhao. "Research on Heart Sound Classification Based on SVM Model." In **2024 4th International Conference on Electronic Information Engineering and Computer Communication (EIECC)**, pp. 497–501. IEEE, 2024. [Online]. Available: <https://doi.org/10.1109/EIECC64539.2024.10929207>
- [16] Mishra, Ansh, Jia Qi Yip, and Eng Siong Chng. "Time-Domain Heart Sound Classification Using Neural Audio Codecs." In **2024 11th International Conference on Advanced Informatics: Concept, Theory and Application (ICAICTA)**, pp. 1–5. IEEE, 2024. [Online]. Available: <https://doi.org/10.1109/ICAICTA63815.2024.10763239>
- [17] Li, Xin, G. Andre Ng, and Fernando S. Schlindwein. "Transfer Learning in Heart Sound Classification using Mel Spectrogram." In **2022 Computing in Cardiology (CinC)**, pp. 1–4. IEEE, 2022. [Online]. Available: <https://doi.org/10.22489/CinC.2022.046>
- [18] Grooby, Ethan, C. Sitaula, D. Fattahi, Reza Sameni, K. Tan, L. Zhou, A. King, A. Ramanathan, A. Malhotra, Guy A. Dumont, and Faezeh Marzbanrad. "Real-Time Multi-Level Neonatal Heart and Lung Sound Quality Assessment for Telehealth Applications." In **IEEE Access**, vol. 10, pp. 10934–10948. IEEE, 2022. [Online]. Available: <https://doi.org/10.1109/ACCESS.2022.3144355>